

P0401R3

Providing size feedback in the Allocator interface

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Abstract

Utilize size feedback from Allocator to reduce spurious reallocations

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§ 1. Introduction

This is a library paper to describe how size feedback can be used with allocators:

- In the case of `std::allocator`, the language feature proposed in [\[P0901R5\]](#) could be used.
- In the case of custom allocators, the availability of the traits interface allows standard library containers to leverage this functionality.

§ 2. Motivation

Consider code adding elements to `vector`:

```
std::vector<int> v = {1, 2, 3};  
// Expected: v.capacity() == 3  
  
// Add an additional element, triggering a reallocation.  
v.push_back(4);
```

Many allocators only allocate in fixed-sized chunks of memory, rounding up requests. Our underlying heap allocator received a request for 12 bytes ($3 * \text{sizeof}(\text{int})$) while constructing `v`. For several implementations, this request is turned into a 16 byte region.

§ 2.1. Why not `realloc`

`realloc` poses several problems:

- We have unclear strategies for growing a container. Consider the two options (a and b) outlined in the comments of the example below. Assume, as with the [§2 Motivation](#) example, that there is a 16 byte size class for our underlying allocator.

```
std::vector<int> v = {1, 2, 3};  
// Expected: v.capacity() == 3  
  
// Add an additional element. Does the vector:  
// a) realloc(sizeof(int) * 4), adding a single element  
// b) realloc(sizeof(int) * 6), attempt to double, needing reallocation  
v.push_back(4);  
  
// Add another element.  
// a) realloc(sizeof(int) * 5), fails in-place, requires reallocation  
// b) size + 1 <= capacity, no reallocation required  
v.push_back(5);
```

Option a requires every append (after the initial allocation) try to `realloc`, requiring an interaction with the allocator each time.

Option b requires guesswork to probe the allocator's true size. By doubling (the typical growth pattern for `std::vector` in `libc++` and `libstdc++`), we skip past the existing allocation's size boundary. While we can use a reduced growth rate, doing so requires more interactions with the allocator (calls to `realloc`), as well as copies (when we cannot resize in-place).

Both of these options add extra branching on the control flow for growing the buffer.

- When the allocator cannot resize the buffer in-place, it allocates new memory and `memcpy`'s the contents. For non-trivially copyable types, this is disastrous. The *language* lacks support for reallocating in-place *without* moving. While this function could be implemented as magic library function, it could not interact with many underlying `malloc` implementations (`libc`, in the case of `libc++` and `libstdc++`) used to build `operator new`, nor could it interact with user-provided replacement functions.

In-place reallocation has been considered in [\[N3495\]](#) (throwing `std::bad_alloc` when unable) and [\[P0894R1\]](#) (returning `false` when unable). This has sometimes been referred to as an "extend" or an "expand" interface. Extending an existing allocation requires a round-trip to the allocator (and its requisite branch/call overheads) to determine whether the allocation *could* be expanded. As discussed previously, we need to probe for the true size of the backing object (until we find extend fails) until we give up and apply a multiplicative expansion factor.

For fixed-size allocators it makes more sense, and is much simpler for containers to adapt to, if the allocator is able to over-allocate on the initial request and inform the caller how much memory was made available.

§ 2.2. Why not ask the allocator for the size

We could also explore APIs that answer the question: "If I ask for `N` bytes, how many do I actually get?" [\[jemalloc\]](#) and [\[TCMalloc\]](#) call this `nallocx`.

- `nallocx` requires an extra call / size calculation. This requires we duplicate work, as the size calculation is performed as part of allocation itself.
- `nallocx` impairs telemetry for monitoring allocation request sizes. If a `nallocx` return value is cached, the user appears to be asking for exactly as many bytes as `nallocx` said they would get.

See also [\[P0901R5\]](#)'s discussion "`nallocx`: not as awesome as it looks."

§ 3. Proposal

Wording relative to [\[N4800\]](#).

We propose a free function to return the allocation size simultaneously with an allocation.

Amend [allocator.requirements]:

```
template<typename Pointer>
struct allocation_result {
    Pointer ptr;
    size_t count;
};
```

The class template `allocation_result` has the template parameters, data members, and special members specified above. It has no base classes or members other than those specified.

```
template<typename Allocator>
constexpr allocation_result<Allocator::pointer> allocate_at_least(Allocator& a, size_t n)
{
};
```

- *Effects:* Let `s` be the result of `allocate_at_least(a, n)`. Memory is allocated for at least `n` objects of type `T` but objects are not constructed.
- *Returns:* `allocation_result{ptr, m}`, where `ptr` is that memory and `m` is the number of objects for which memory has been allocated, such that `m >= n`.
- [*Note:* If `s.count == 0`, the value of `s.ptr` is unspecified. -- end note]

Table 34 - Cpp17Allocator Requirements

<code>a.allocate_at_least(n)</code>	<code>allocation_result<X::pointer></code>	Returns <code>{a.allocate(n), n}</code> Memory is allocated for at least <code>n</code> objects of type <code>T</code> but objects are not constructed and returned as <code>ptr</code>. The actual number of objects <code>m</code>, such that <code>m >= n</code>, that memory has been allocated for is returned in the second value. <code>allocate_at_least</code> may throw an appropriate exception. [<i>Note:</i> If <code>n == 0</code>, the value of <code>ptr</code> is unspecified. — end note]
<code>a.deallocate(p,n)</code>	(not used)	<i>Preconditions:</i> <code>p</code> shall be a value returned by an earlier call to <code>allocate</code> or <code>allocate_at_least</code> that has not been invalidated by an intervening call to <code>deallocate</code>. If this memory was obtained by a call to <code>allocate</code>, <code>n</code> shall match the value passed to <code>allocate</code> to obtain this memory. Otherwise, <code>n</code> shall satisfy <code>capacity >= n >= requested where [p, capacity] = allocate_at_least(requested)</code> was used to obtain this memory. <i>Throws:</i> Nothing.

Amend [allocator.members]

Except for the destructor, member functions of the default allocator shall not introduce data races as a result of concurrent calls to those member functions from different threads. Calls to these functions that allocate or deallocate a particular unit of storage shall occur in a single total order, and each such deallocation call shall happen before the next allocation (if any) in this order.

```
[[nodiscard]] constexpr T* allocate(size_t n);
```

- *Returns:* A pointer to the initial element of an array of storage of size `n * sizeof(T)`, aligned appropriately for objects of type `T`.
- *Remarks:* the storage is obtained by calling `::operator new`, but it is unspecified when or how often this function is called.
- *Throws:* `bad_array_new_length` if `SIZE_MAX / sizeof(T) < n`, or `bad_alloc` if the storage cannot be obtained.

```
[[nodiscard]] constexpr allocation_result<T*> allocate_at_least(size_t n);
```

- *Returns:* `allocation_result{ptr, count}`, where `ptr` is a pointer to the initial element of an array of storage of size `count * sizeof(T)`, aligned appropriately for objects of type `T`, and `count >= n`.
- *Remarks:* the storage is obtained by calling `::operator new`, but it is unspecified when or how often this function is called.
- *Throws:* `bad_array_new_length` if `SIZE_MAX / sizeof(T) < n`, or `bad_alloc` if the storage cannot be obtained.

```
constexpr void deallocate(T* p, size_t n);
```

- *Preconditions:* `p` shall be a pointer value obtained from `allocate()` or `allocate_at_least`. If this memory was obtained by a call to `allocate`, `n` shall equal the value passed as the first argument to the invocation of `allocate` which returned `p`. Otherwise, `n` shall satisfy `count >= n >= requested` where `[p, count] = allocate_at_least(requested)` was used to obtain this memory.
- *Effects:* Deallocates the storage referenced by `p`.
- *Remarks:* Uses `::operator delete`, but it is unspecified when this function is called.

§ 4. Design Considerations

§ 4.1. `allocate` selection

There are multiple approaches here:

- Return a pointer-size pair, as presented.
- Overload `allocate` and return via a reference parameter. This potentially hampers optimization depending on ABI, as the value is returned via memory rather than a register.

§ 4.2. Size Return Value

In [Prague], LEWG discussed the return value. For compatibility with the existing allocator APIs, which work in units of *objects* rather than bytes, this proposal chooses to continue to return an integer number of objects.

Additionally, for types with non-trivial alignment requirements, we must allocate storage for objects, rather than bytes, as raw bytes do not convey the appropriate alignment needs to the allocator.

For example: In the `std::vector<T>` case, many implementations use 3 pointers for representing the state of the container (begin, end, and capacity). If we preserved the precise value returned by the underlying allocator, we may not be able to legally form the capacity pointer. For these implementations, replacing the capacity pointer with a capacity in bytes would be an ABI break.

§ 4.3. deallocate changes

We now require flexibility on the size we pass to `deallocate`. For container types using this allocator interface, they are faced with the choice of storing *both* the original size request as well as the provided size (requiring additional auxillary storage), or being unable to reproduce one during the call to `deallocate`.

As the true size is expected to be useful for the `capacity` of a `string` or `vector` instance, the returned size is available without additional storage requirements. The original (requested) size is unavailable, so we relax the `deallocate` size parameter requirements for these allocations.

§ 4.4. Interaction with `polymorphic_allocator`

`std::pmr::memory_resource` is implemented using virtual functions. Adding new methods, such as the proposed allocate API would require taking an ABI break.

§ 4.5. Zero-Sized Requests

In Prague, LEWG discussed the behavior of `allocate_at_least(allocator, 0)`. This maximizes implementation freedom.

- The return value may not be dereferenceable (even if not `nullptr`).
- Other implementations, such as a stack allocator, could just return the current offset.

§ 5. Revision History

§ 5.1. R2 → R3

Applied LEWG feedback from [\[Prague\]](#).

- Renamed `sized_ptr_t` to `allocation_result`, based on discussion in Prague that this library proposal (which returns a *count* of storage for objects) have a distinct type from the underlying language proposal ([\[P0901R5\]](#), which returns bytes). This was one of the names bikeshedded during the review of P0901R5 ([\[P0901R5-Review\]](#)).
- Renamed `n` (the returned size) to `count`, based on discussion in Prague to clarify that is measured in objects, rather than bytes. Additionally, augmented discussion of the `deallocate` changes for why implementations may not be able to preserve the precise size returned by the underlying allocator.

Poll: We prefer to return number of bytes.

SF	F	N	A	SA
0	0	6	6	5

- Changed the template parameter to pointer type to allow for fancy pointers.

POLL: Change the template parameter to `sized_ptr_t` to pointer type (to support fancy ptr).

Unanimous consent

- Added design intent and wording that `allocation_result` be compatible with structured bindings. This wording was adapted from the proposed resolution of [\[LWG3373\]](#).
- Added `constexpr`
- Removed default template parameter from `allocation_result`
- Based on LEWG feedback, changed `allocate_at_least` to deduce return type from the allocator. Since the pointer type is chosen by the allocator, the underlying object type is less relevant. This also removed the default template parameter.

POLL: We want to deduce the object type from the allocator (`alloc_least_n` [sic]).

SF	F	N	A	SA
4	8	1	0	0

POLL: We like the design, please revise and do initial wording review (with Pablo, and David Stone) send to LWG (tentatively Ready).

SF	F	N	A	SA
6	12	0	0	0

§ 5.2. R1 → R2

Applied LEWG feedback from [\[Cologne\]](#).

Poll: We want a feature like this.

SF	F	N	A	SA
2	9	2	0	0

- As users may specialize `std::allocator_traits`, this revision moves to a free function `std::allocate_at_least` as suggested via a LEWG poll.

Poll: Name choice

5	<code>allocate_with_size</code>
4	<code>overallocate</code>
2	<code>allocate_with_oversubscribe</code>
6	<code>allocate_oversize</code>
14	<code>allocate_at_least</code>

Poll: Prefer a struct rather than a requirement of structured bindings.

SF	F	N	A	SA
2	8	0	2	1

§ References

§ Informative References

[Cologne]

[Cologne Meeting Minutes](#). 2019-07-17. URL: <http://wiki.edg.com/bin/view/Wg21cologne2019/P0401>

[JEMALLOC]

[jemalloc\(3\) - Linux man page](#). URL: <http://jemalloc.net/jemalloc.3.html>

[LWG3373]

[{to,from}_chars_result and format_to_n_result need the 'we really mean what we say' wording](#). URL: <https://cplusplus.github.io/LWG/lwg-active.html#3373>

[N3495]

Ariane van der Steldt. [inplace realloc](#). 7 December 2012. URL: <https://wg21.link/n3495>

[N4800]

[Working Draft, Standard for Programming Language C++](#). 2019-01-21. URL: <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2019/n4800.pdf>

[P0894R1]

[realloc for C++](#). 2019-01-18. URL: <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2019/p0894r1.md>

[P0901R5]

[Size feedback in operator new](#). 2019-10-03. URL: <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2019/p0901r5.html>

[P0901R5-Review]

[P0901R5 LEWG Review Prague](#). 2020-02-10. URL: <http://wiki.edg.com/bin/view/Wg21prague/P0901>

[Prague]

[P0401R2 LEWG Review Prague](#). 2020-02-14. URL: <http://wiki.edg.com/bin/view/Wg21prague/P0401>

[TCMalloc]

[TCMalloc](#). URL: <https://github.com/google/tcmalloc>